

Dermatological Disorders: Acne

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INTRODUCTION

- There are Several skin disorders. Some patients seek the
 - Assistance of a health care provider
 - Utilize self-care.
- This **acne vulgaris**, **contact dermatitis** (irritant and allergic), and **diaper dermatitis**; other **common skin** and **soft tissue infections** and **superficial fungal infection**

Acne vulgaris

- Acne is a common, usually self-limiting dx involving inflammation of the sebaceous follicles of the face and upper trunk.
- Acne is a chronic skin disease characterized by
 - **Open or closed comedones** and
 - **Inflammatory lesions**, including papules, pustules, and nodules (cysts).



Papulopustular acne, some inflammatory papules become nodular and thus represent early stages of nodulocystic acne.

Pathophysiology

- Acne usually begins in the **prepubertal period** and progresses as **androgen production** and **sebaceous gland activity increase** with gonad development.
- Acne is a multifactorial disease. **Genetic, racial, hormonal, dietary, and environmental factors** have been implicated in its development. Its **psychologic impact** can be severe.
- Acne progresses through four stages:
 1. Increased follicular keratinization,
 2. Increased sebum production,
 3. Bacterial lipolysis of sebum triglycerides to free fatty acids-
Cutibacterium (***Propionobacterium***) ***Acnes***
 4. Inflammation

Pathophysiology...

- Circulating androgens cause sebaceous glands to increase their size and activity. There is increased keratinization of epidermal cells and development of an obstructed sebaceous follicle, called a *microcomedone*.
- Cells adhere to each other, forming a **dense keratinous plug**.
- **Sebum**, produced in increasing amounts, becomes trapped behind the keratin plug and solidifies, contributing to **open** or **closed comedone formation**.
- Pooling of sebum in the follicle facilitates proliferation of the anaerobic bacterium ***P.acnes***, which generates a T-cell response resulting in **inflammation**.

Pathophysiology...

- ***P. acnes*** produces a lipase that hydrolyzes sebum triglycerides into free fatty acids that may increase keratinization and lead to microcomedone formation.
- The **closed comedone** (whitehead) is the first visible lesion of acne. It is almost completely obstructed to drainage and has a tendency to rupture.
- An **open comedone** (blackhead) is formed as the plug extends to the upper canal and dilates its opening. Acne characterized by open and closed comedones is termed ***noninflammatory acne***



Closed Comedones

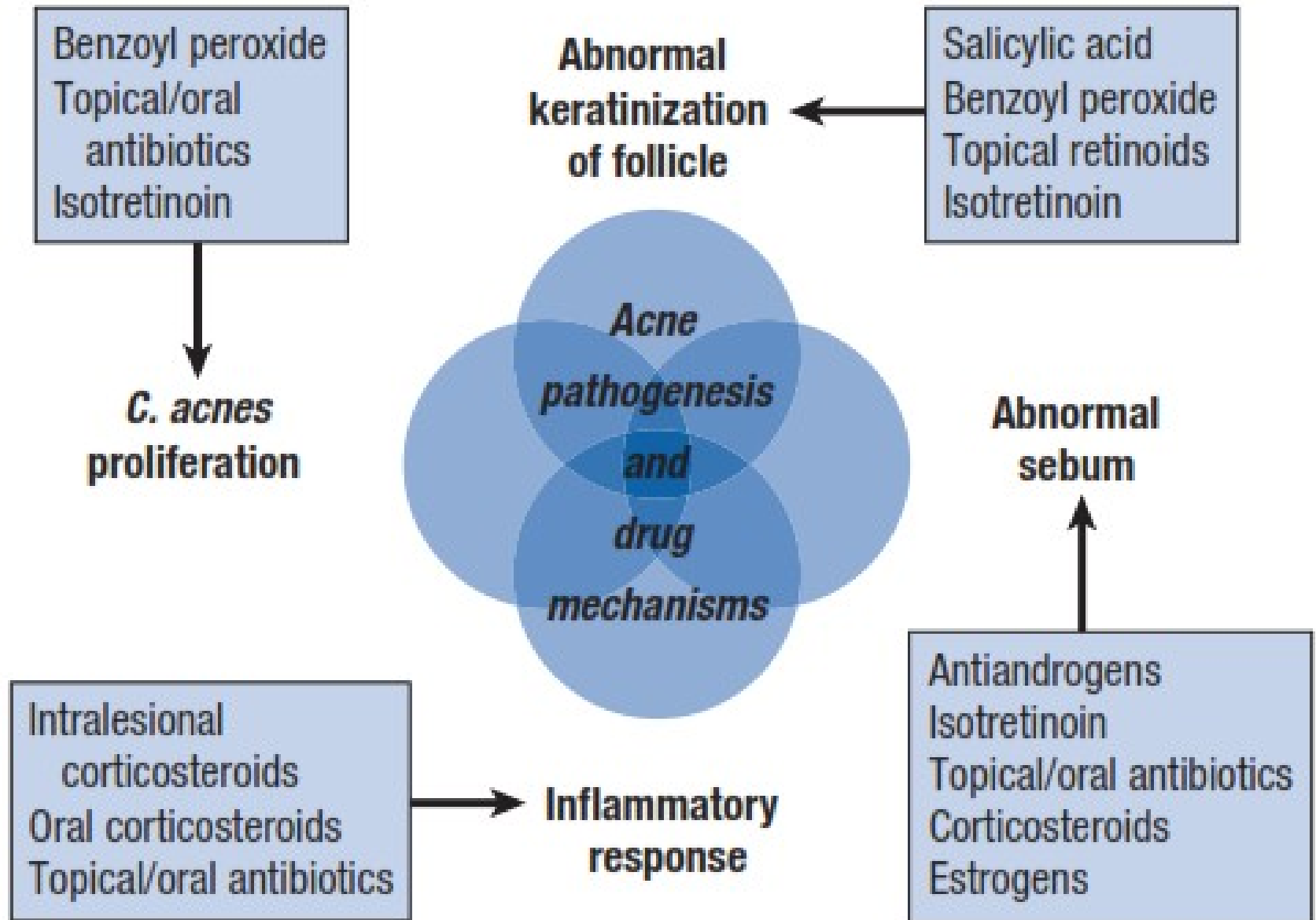


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Open comedones

- **Pus formation** occurs due to recruitment of neutrophils into the follicle during the inflammatory process and release of *P. acnes*–generated chemokines.
- *P. acnes* also produces enzymes that increase permeability of the follicular wall, causing it to rupture, thereby releasing **keratin, lipids, and irritating free fatty acids into the dermis.**
- Inflammatory lesions that may form and lead to scarring include pustules, nodules, and cysts.

Acne pathogenesis and drug mechanisms



CLINICAL PRESENTATION

- **Lesions** usually occur on the face, back, upper chest, and shoulders.
- Severity varies from a mild comedonal form to severe inflammatory acne.
- The disease is categorized as mild, moderate, or severe, depending on the type and severity of lesions.
- Lesions may take months to heal completely, and fibrosis associated with healing may lead to permanent scarring.

Treatment of Acne Vulgaris

Goals of Treatment:

- To reduce the number and severity of lesions,
- slow disease progression,
- limit disease duration,
- Improve appearance
- Prevent formation of new lesions,
- Prevent scarring and hyperpigmentation and
- Avoid psychologic suffering.

GENERAL APPROACH

Global Alliance to Improve Outcomes in Acne consensus statements:

- ➡ Acne should be approached as a chronic disease.
- ➡ Strategies to limit antibiotic resistance are important in acne management.
- ➡ Combination retinoid-based therapy is first-line therapy.
- ➡ Topical retinoids should be first-line agents in maintenance therapy.
- ➡ Early, appropriate treatment is best to minimize potential for acne scars.
- ➡ Adherence should be assessed via verbal interview or use of a simple tool.

Nonpharmacologic Therapy

- Encourage pts to avoid aggravating factors, maintain a balanced diet, and control stress.
- Pts should wash no more than twice daily with a mild, nonfragranced opaque or glycerin soap or a soapless cleanser.
- Scrubbing should be minimized to prevent follicular rupture.
- Comedone extraction results in immediate cosmetic improvement but has not been widely tested in clinical trials.

Pharmacologic Therapy

- ***Comedonal noninflammatory acne:*** Select topical agents that target the increased keratinization by producing exfoliation.
 - **Topical retinoids** (especially adapalene) are drugs of choice.
 - Benzoyl peroxide or azelaic acid can be considered.
- ***Mild to moderate papulopustular inflammatory acne:*** It is important to reduce the population of *P. acnes*.
 - Either the fixed-dose combination of adapalene and benzoyl peroxide or the fixed-dose combination of topical clindamycin and benzoyl peroxide is first choice therapy

- As alternatives, a different **topical retinoid** used with a different topical antimicrobial agent could be used, with or without benzoyl peroxide. Azelaic acid or benzoyl peroxide can also be recommended.
- ***Severe papulopustular or moderate nodular acne:***
 - Oral isotretinoin monotherapy is first choice.
 - Alternatives include systemic antibiotics in combination with adapalene, with the fixed-dose combination of adapalene and benzoyl peroxide or in combination with azelaic acid.
 - consider oral antiandrogens in combination with oral antibiotics or topical treatments, or systemic antibiotics in combination with benzoyl peroxide.

- ***Nodular or conglobate acne:***

- Monotherapy with oral isotretinoin is first choice.
- An alternative is systemic antibiotics in combination with azelaic acid.
- Consider oral antiandrogens in combination with oral antibiotics, systemic antibiotics in combination with adapalene, benzoyl peroxide, or the adapalene-benzoyl peroxide fixed-dose combination.

- ***Maintenance therapy for acne:***

- Topical retinoids are most commonly recommended (adapalene, tazarotene, or tretinoin). Topical azelaic acid is an alternative.
- Maintenance is usually begun **after a 12-week** induction period and continues for **3 to 4 months**. A longer duration may be necessary to prevent relapse upon discontinuation. Long-term therapy with antibiotics is not recommended to minimize antibiotic resistance.

Acne Severity	MILD MODERATE SEVERE				
	Comedonal	Mixed and Papular/pustular	Mixed and Papular/pustular	Nodular ⁽²⁾	Nodular/Conglobate
1st Choice	Topical Retinoid	Topical Retinoid + Topical Antimicrobial	Oral Antibiotic + Topical Retinoid +/- BPO	Oral Antibiotic + Topical Retinoid + BPO	Oral Isotretinoin ⁽²⁾
Alternatives ⁽¹⁾	Alt. Topical Retinoid or Azelaic acid* or Salicylic acid	Alt. Topical Retinoid Antimicrobial Agent + Alt. Topical Retinoid or Azelaic Acid*	Alt. Oral Antibiotic + Alt. Topical Retinoid +/- BPO	Oral Isotretinoin or Alt. Oral Antibiotic + Alt. Topical Retinoid	High Dose Oral Antibiotic + Topical Retinoid + BPO
Alternatives for Females ^(1,4)	See 1st Choice	See 1st Choice	Oral Antiandrogen + Topical Retinoid/ Azelaic Acid* +/- Topical Antimicrobial	Oral Antiandrogen ⁽³⁾ + Topical Retinoid/ +/- Oral Antibiotic +/- Alt. Antimicrobial	High Dose Oral Antiandrogen ⁽³⁾ + Topical Retinoid +/- Alt. Topical Antimicrobial
Maintenance Therapy	Topical Retinoid		Topical Retinoid +/- BPO		

- Exfoliants induce continuous mild drying and peeling by irritation, damaging superficial skin layers and inciting inflammation.
- **Resorcinol (2%, 3%)** is less keratolytic than **salicylic acid**
- **Salicylic acid (2%-10%)** is keratolytic, has mild antibacterial activity against *P. acnes*, and offers slight antiinflammatory activity at concn up to 5%.
- **Sulfur (2%-10%)** is keratolytic and has antibacterial activity.

Topical Retinoids

- Retinoids reduce obstruction within the follicle and are useful for both comedonal and inflammatory acne.
- They reverse abnormal keratinocyte desquamation and are active keratolytics. They inhibit microcomedone formation, decreasing the number of mature comedones and inflammatory lesions.
- Topical retinoids are safe, effective, and economical for treating all but the most severe cases of acne. They should be the first step in moderate acne, alone or in combination with antibiotics and benzoyl peroxide.
- Side effects include erythema, xerosis, burning, and peeling.

- **Tretinoin** (retinoic acid and vitamin A acid) is available as 0.05% solution (most irritating); 0.01% and 0.025% gels; and 0.025%, 0.05%, and 0.1% creams (least irritating).
- **Adapalene** is the topical retinoid of first choice for both treatment and maintenance therapy because it is as effective but less irritating. It is available as 0.1% gel, cream, alcoholic solution, and pledgets. A 0.3% gel formulation is also available.
- **Tazarotene** is as effective as adapalene in reducing noninflammatory and inflammatory lesion counts when applied half as frequently. Compared with tretinoin, it is as effective for comedonal and more effective for inflammatory lesions when applied once daily. The product is available as a 0.05% and 0.1% gel or cream.

- **Benzoyl peroxide** is bactericidal and also suppresses sebum production and reduces free fatty acids. It is useful for both noninflammatory and inflammatory acne. It has a rapid onset and may decrease the number of inflamed lesions within 5 days.
- Used alone or in combination, benzoyl peroxide is the standard of care for mild to moderate papulopustular acne. It is often combined with topical retinoids or an antimicrobial.

- For maintenance therapy, benzoyl peroxide can be added to a topical retinoid.
- Soaps, lotions, creams, washes, and gels are available in concentrations of 1% to 10%.
- Alcohol-based gel preparations generally cause more dryness and irritation.
- **Topical erythromycin** and **clindamycin** have become less effective due to resistance by *P. acnes*.

- Addition of BPO or topical retinoids to the macrolide is more effective than antibiotic monotherapy.
- Clindamycin is preferred because of potent action and lack of systemic absorption.
- **Azelaic acid** has antibacterial, antiinflammatory, and comedolytic activity. It is used for mild to moderate inflammatory acne but has limited efficacy compared with other therapies.
- **Dapsone** 5% topical gel is a sulfone that has anti-inflammatory and antibacterial properties that improve both inflammatory and noninflammatory acne.

- Systemic antibiotics are standard therapy for moderate and severe acne and treatment-resistant inflammatory acne.
- **Erythromycin** is effective, but because of bacterial resistance, its use should be limited to patients who cannot use a tetracycline derivative (eg, pregnant women and children <8 years old).
- **Ciprofloxacin, cotrimoxazole and trimethoprim** alone are also effective
- **Tetracyclines (minocycline and doxycycline)** have antibacterial and anti-inflammatory effects.

- **Isotretinoin** decreases sebum production, inhibits *P. acnes* growth, and reduces inflammation.
- Isotretinoin is the only drug treatment for acne that produces prolonged remission.
- The approved dose is 0.5 to 2 mg/kg/day, usually given over a 20-week course. Drug absorption is greater when taken with food.
- Initial flaring can be minimized by starting with 0.5 mg/kg/day or less. Alternatively, lower doses can be used for longer periods, with a total cumulative dose of 120 to 150 mg/kg.

- **Oral contraceptives** containing estrogen can be useful for acne in some women. Agents with FDA approval for this indication include **norgestimate with ethinyl estradiol** and **norethindrone acetate with ethinyl estradiol**; other estrogen-containing products may also be effective.
- **Spironolactone** in higher doses is an antiandrogenic compound. Doses of 50 to 200 mg have been shown to be effective in acne.
- **Cyproterone acetate** is an antiandrogen that may be effective for acne in females when combined with ethinyl estradiol (in the form of an oral contraceptive).
- **Oral corticosteroids** in high doses used for short courses may be of temporary benefit in patients with severe inflammatory acne.

EVALUATION OF THERAPEUTIC OUTCOMES

- Patient should record the objective response to treatment in a diary.
- Contact patients within 2 to 3 weeks after the start of therapy to assess progress.
- Lesion counts should decrease by 10% to 15% within 4 to 8 weeks or by more than 50% within 2 to 4 months.
- Inflammatory lesions should resolve within a few weeks, and comedones should resolve by 3 to 4 months.
- Long-term parameters should include no progression of severity, lengthening of acne-free periods throughout therapy, and no further scarring or pigmentation throughout therapy.
- Monitor patients regularly for adverse treatment effects,